

FILTERING-QUERIES

1. Filtering by node labels.

Give all the people who worked in the movie called 'The Matrix'

```
MATCH (p:Person)-[:ACTED_IN]->(m:Movie)
WHERE m.title='The Matrix'
RETRUN p.name
```

2.Filtering Using ranges

Find the movies and person nodes where the person acted in a movie that was released in 2008 or 2009

```
MATCH (p:Person)-[:ACTED_IN]->(m:Movie)
WHERE m.released = 2008 or m.released = 2009
RETRUN p.name
```

Retrieve person name, movie title, movie released where movies released between 2000 and 2003

```
MATCH (p:Person)-[:ACTED_IN]->(m:Movie)
WHERE m.released <= 2003 AND m.released >= 2000
RETRUN p.name,m.title,m.released
```

3. Filtering by existence of a property

Return only movie nodes where Jack Nicholson acted in the movie and movie has tagline property

```
MATCH (p:Person)-[:ACTED_IN]->(m:Movie)
WHERE p.name = 'Jack Nicholson' AND m.tagline IS NOT NULL
RETRUN m.title, m.tagline
```

4. Filtering with partial strings

Find all actors in the graph where first name is Michael

```
MATCH (p:Person)-[:ACTED_IN]->(m:Movie)
WHERE p.name STARTS WITH 'Michael'
RETRUN p.name
```

5.Filtering by patterns in the graph

Find all the people who wrote a movie but did not direct that same movie

```
MATCH (p:Person)-[:WROTE]->(m:Movie)
WHERE NOT EXISTS ( (p)-[:DIRECTED]-(m))
RETRUN p.name,m.title
```

6. Filtering with existing lists in the graph

Name of the actor who played role Neo in the movie The Matrix

```
MATCH (p:Person)-[r:ACTED_IN]->(m:Movie)
WHERE 'Neo' in r.roles AND m.name = 'The Matrix'
RETRUN p.name, r.roles
```

7.What properties does a node or relationship have?

List of all property keys for a node

```
MATCH (p:Person)
RETURN p.name, keys(p)
```

8. What properties exist in the graph

```
CALL db.propertykeys()
```

Challenges

1. Retrieve all movies that have a released property value in 2000,2002,2004,2006,or 2008.

```
MATCH (m:Movie) WHERE m.released in
[2000,2002,2004,2006,2008] RETURN m.title
```

2. Find all the people who born between 1970 to 1979

```
MATCH (p:Person) WHERE p.born >= 1970 AND p.born < 1979
RETURN p.name
```

3.Select all the actors born after 1960 who acted in the movie title "As Good as It Gets"

```
MATCH (p:Person)-[:ACTED_IN]->
(m:Movie {title: 'As Good as It Gets'})
WHERE p.born >= 1960 RETURN
p.name
```